Marco Badran

Postdoctoral researcher at Bocconi University

email | website

October 30, 2026

Employment

Postdoctoral researcher at Bocconi University (BIDSA)
2026 – ongoing
Mentor: Prof. Alessandro Pigati.

Postdoctoral researcher at ETH Zürich
2023 – 2026
Mentor: Prof. Joaquim Serra.

Education

Ph.D. in Mathematics
University of Bath, 2019 – 2023
Thesis Title: *Concentrating solutions in the magnetic Ginzburg–Landau model.*
Supervisor: Prof. Manuel del Pino.

Master in Mathematics
Università statale di Milano, 2017 – 2019
Thesis Title: *The Allen–Cahn equation: a conjecture by De Giorgi.*
Supervisor: Prof. Bernhard Ruf.

Bachelor in Mathematics
Università statale di Milano, 2014 – 2017

Preprints and publications

- *Nonabelian Higgs fields from minimal submanifolds.* **arXiv**.
- *Stable solutions in the abelian Higgs model.* **arXiv**.
- *A two-dimensional Allen–Cahn theory for interfaces with boundary.* (with M. del Pino and M. A. M. Guaraco) **arXiv**.
- *Uniform small energy regularity for fractional geometric problems.* (with G. Cozzi) **arXiv**.
- *Nonlocal free boundary minimal surfaces.* (with S. Dipierro and E. Valdinoci) Accepted in **Ann. Sc. Norm. Super. Pisa Cl. Sci. arXiv**.
- *Stationary p-harmonic maps approaching planar singular harmonic maps to the circle.* (with J. Van Schaftingen) **Calc. Var. PDE**.
- *Harmonic maps to the circle with higher dimensional singular set.* **Proc. London Math. Soc**.

- *Solutions of the Ginzburg–Landau equations concentrating on codimension-2 minimal submanifolds.* (with M. del Pino) **J. London Math. Soc**.
- *Entire solutions to 4-dimensional Ginzburg–Landau equations and codimension 2 minimal submanifolds.* (with M. del Pino) **Adv. Math**.

Invited talks and seminars

- Centro De Giorgi seminar, Pisa (07.26)
- Regularity Theory for Free Boundary and Geometric Variational Problems VI, Cervia (06.26)
- London Analysis and Probability Seminar, King's College London (04.26)
- Analysis seminar, Politecnico di Torino (12.25)
- La Cristalera PDE days, Madrid (11.25)
- International geometric analysis conference in Milan, Bocconi (06.25)
- PDE Seminar, Cambridge (05.25)
- Analysis Seminar, SNS Pisa (03.25)
- Analysis Seminar, ETH Zürich (02.25)
- Analysis & PDE seminar, NUS Singapore (02.24)
- Young researchers in PDEs, UAM/ICMAT Madrid (10.23)
- Analysis seminar, University of Bath (02.23)
- Analysis seminar, UAM/ICMAT Madrid (01.23)
- Geometry seminar, UCL London (01.23)
- Analysis seminar, UniMi Milano (12.22)

Teaching

- Co-lecturer of *Geometric Heat Flows*, D-MATH (Fall 2025)
- Organiser for *Mathematik I*, D-USYS ETHZ (Fall 2025)
- Organiser for *Partial Differential Equations*, D-MATH ETHZ (Spring 2025)
- Organiser for *Analysis I*, D-INFK ETHZ (Spring 2025)
- Lecturer for *Nonlinear Analysis and Perturbation Methods*, D-MATH ETHZ (Fall 2024)
- Organiser for *Analysis IV*, D-MATH ETHZ (Spring 2024)
- Graduate Teaching Assistant at University of Bath (several courses, 2019 – 2023)

Other activities

Referee for: Adv. Differential Equations, Ars Inven. Analytica, Proc. London Math. Soc., Discrete Contin. Dyn. Sys., Journal of Differential Equations

Supervision:

- André Dabrowski (Master Thesis, joint with J. Serra, ETH Zürich, 2025)
- Martin Gonzalez Yu (Semester paper – ETH Zürich, 2025)
- Derk Steffens (Master Thesis, joint with J. Serra, ETH Zürich, 2025)
- Marco De Rivo (Bachelor thesis – ETH Zürich, 2025)
- Derk Steffens (Reading Course – ETH Zürich, 2025)
- Luana Jost (Semester paper – ETH Zürich, 2025)
- Lennard Trautmann (Semester paper – ETH Zürich, 2025)
- Matteo Gatti (Bachelor thesis – ETH Zürich, 2025)